Why Separable Subjects Still Fail to Link: A Streaming Journal over Knesset Committee Protocols

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Abstract

We build a longitudinal journal over a stream of Israeli Knesset Finance Committee transcripts, coupling three tasks — topic segmentation, *streaming* cross-document subject linking, and update summarization — under an asymmetric cost model in which merging two distinct matters is worse than splitting one. Our central result is negative and, we argue, the most useful thing the project produced. On a manually annotated set of 523 segments, one fixed representation (`multilingual-e5-large`+ABTT) scores F_1=0.619 judging subject identity over *pairs*, and F_1=0.125 when the *same representation* makes the *same decision* online against a growing journal. The collapse is caused not by weak Hebrew embeddings but by the streaming decision structure, where only 6.9% of arrivals are true repeats. Under a matched streaming-honest protocol a distinctiveness *margin* rule raises F_1 0.087→0.190, while every LLM verifier stays ≤ 0.111. We further document two ways this benchmark rewards look-ahead.

1 Introduction

A parliamentary committee returns to the same matter — a bill, a budget transfer, a programme — again and again over months, so reconstructing the history of any one matter means reading dozens of unrelated transcripts. We produce that history automatically: an index in which each entry is one underlying matter carrying a dated log of its progress. The problem decomposes into three coupled tasks. **Segmentation** partitions each protocol into subject spans; **streaming subject linking** decides, for each arriving span, whether it continues an existing entry or opens a new one; and **log writing** generates only the *incremental* update to the matched entry.

How this is done today, and its limits. Task 2 is a variant of cross-document event coreference (Cybulska and Vossen, 2014; Barhom et al., 2019), but the standard setting is *offline*: a system sees the whole collection and clusters it globally. Ours is online over a growing index, with no look-ahead and its own past errors persisting. Task 3 is update summarization (Dang and Owczarzak, 2008) with retrieval augmentation (Lewis et al., 2020); unlike query-conditioned RAG, faithfulness is defined relative to a *chain of prior summaries* rather than a source document, because the retrieved log records what has already been written.

Asymmetric cost. Merging two distinct matters corrupts an entry permanently and silently; splitting one is visible and repairable. We therefore report every operating point twice: at best F_1, and at the *anti-duplication* point where the false-merge rate (FMR) is held $\leq 5\%$.

Contributions. We expected the bottleneck to be Hebrew representation quality; it is not. Our research question became: *why does near-ceiling subject separability fail to produce usable streaming linking, and what closes the gap?* We contribute (i) a manual gold standard over 211/213 protocols; (ii) a quantification of the separability/decision gap with a base-rate explanation; (iii) a matched ablation of the levers that close it, all structural rather than representational; (iv) evidence that an LLM helps as an *extractor* but not a *classifier*; and (v) two look-ahead leaks found in our own pipeline. Any index built incrementally over a stream faces the same arithmetic, and our diagnosis predicts a better encoder will not help.

2 Data

We use 213 Hebrew transcripts of the 25th Knesset Finance Committee (2022-11-21 to 2023-10-26); each is one meeting, given as committee, ISO date and the full transcript. Protocol-level subject markers yield 151 canonical topics after fuzzy variant merging, but these describe the *meeting*, not the spans inside it, so we manually segmented 211/213 protocols into subject spans labelled with the matter under discussion (Table 1).

Two properties dominate what follows. We deliberately did *not* merge near-duplicate labels across protocols — under our cost rule an unmerged duplicate is a cheap false split, whereas a wrong merge would corrupt the gold itself — so recurrence counts are conservative. And, decisively, repeats are *rare*: 36 of 523 arrivals continue an existing matter, a base rate of 6.9%.

3 Methods

Representation. Segments are embedded with `multilingual-e5-large` (Wang et al., 2024), using discussion content only — no headers, agendas, attendees or speaker names. Dense encoders are severely anisotropic (Ethayarajh, 2019): here cross-topic cosine averages $\approx$0.95, leaving a same/different gap of 0.007–0.066. We apply All-But-The-Top (ABTT) (Mu and Viswanath, 2018) — centre, project out the top-k principal directions, re-normalise — and compare against TF-IDF and AlephBERT (Seker et al., 2022).

Two evaluation protocols. **Pairwise** scores all segment pairs and measures *separability*, with oracle access to the full set. **Streaming** presents segments chronologically and scores each against the journal built so far — the actual task.

Levers. An entry is represented either by its *first span, frozen* or by the running *centroid* of its members; the rule is either a flat threshold ($\cos_{(1)} \geq \theta$) or a *distinctiveness margin* ($\cos_{(1)} - \cos_{(2)} \geq \theta$). We vary one lever at a time, CPU-only, holding transform, fitting protocol and growth model fixed.

LLM as verifier, LLM as extractor. Following retrieve-then-verify we shortlist by dense retrieval and ask an LLM to classify *same/related/new*, from a candidate snippet and from its *full accumulated timeline*. Separately we use it as an *extractor* of stable fingerprints (ministries, laws, programmes, request numbers), fusing identifier overlap additively with the geometric margin. Models: `dictalm2` (Shmidman et al., 2024) and `qwen2.5` at 3B/7B.

Streaming-honest fitting. ABTT and whitening estimate parameters *from data*; fitting once over the whole matrix uses segments that have not arrived. We refit at every step from the observed prefix, and report batch and honest numbers side by side.

Log writing. Each update is conditioned on the running journal built from the model's *own* prior generations, so faithfulness is chain-relative. Since predicted links are only $\approx$13% precise, evaluating through them would conflate two failures; we use **gold chains** (22 matters, 58 entries), judged by `Qwen2.5-3B-Instruct` — neither generator — for *faithfulness* and *novelty*.

4 Results

De-anisotropisation matters more than encoder choice. Table 2 shows that removing a *single* principal direction moves `e5` from ARI 0.124 to 0.657 — more than the entire spread across `e5`, TF-IDF and AlephBERT. Before ABTT, TF-IDF beats every dense model: the discriminative signal is lexical and the dense geometry is dominated by shared bureaucratic register. We fix `e5`+ABTT-k1 throughout.

The separability/decision gap. Table 3 and Fig. 1 give the central result: a representation separating same- from different-matter pairs at F_1 0.619 collapses to F_1 0.125 when the same decision is made online. **The bottleneck is the streaming decision structure, not representation quality.**

The cause is a base-rate effect. Only 36 of 523 arrivals are repeats, and each is compared against a journal growing toward several hundred entries, so with n candidates expected false merges scale as $n \cdot \mathrm{FPR}$ while true positives stay at most one — a false-positive rate low enough to look excellent pairwise still floods the true matches. Notably, **retrieval was never the bottleneck**: ranking places the correct entry in the top 10 for *all* 36 repeats (recall@1 0.694, @5 0.944, @10 1.000). The information is in the shortlist; the *decision* fails. Table 7(a) shows why: a segment and its true continuation — same ministry, same request number — score 0.266, while an unrelated ministry's transfer scores 0.448, shared administrative phrasing outweighing the sparse identifying tokens.

What closes the gap. Table 4 varies one lever at a time with everything else fixed. The **distinctiveness margin** is the decisive change: F_1 $0.087 \rightarrow 0.190$ and recall at FMR$\leq 5\%$ from 0% to 8.6%. It targets the failure mode in Table 7(a) — with families of near-identical templates, high absolute similarity is uninformative while a large *gap* is not.

Replacing the frozen first span with the running centroid is a large *recall* fix: on the batch-fit ABTT-$k1$ path it lifts recall $32\% \rightarrow 89\%$ (F_1 $0.103 \rightarrow 0.107$), because the streaming setting keeps supplying evidence and the baseline discards it. We report it on that path because under the matched protocol the threshold rule degenerates to accepting almost everything, leaving the two threshold rows indistinguishable. It improves the entry representation without, on its own, converting into F_1 — a distinction an earlier version of this analysis blurred by combining rows fitted under different protocols.

Extractor, not classifier. As a **verifier** judging from a candidate snippet the best LLM reached F_1 0.111; given the *full timeline* it got *worse* (0.068), the longer context increasing its willingness to merge. As an **extractor** of fingerprints fused with the geometric score it helped: under a common batch-fit setting, geometry alone 0.2000 versus 0.2407 (`dictalm2`) and 0.2013 (`qwen7b`), and a hard identifier gate reaches 11.4% recall at FMR$\leq$ 5% — four times the baseline. It is the natural remedy for Table 7(a).

Two look-ahead leaks. Table 5 treats the fitting protocol as an experimental variable. **(1) Batch-fit transforms.** Whitening needs a 50-dimensional covariance estimate that does not exist early in the stream; refit honestly it collapses $0.250 \rightarrow 0.094$, while cheap transforms survive. **(2) Undefined cold-start margins.** With one entry the runner-up is undefined; a sentinel -1 makes the margin $\cos + 1$, clearing any threshold, so the second segment of every run merges unconditionally. **On streaming benchmarks the fitting protocol must be reported alongside the metric.**

Log writing: faithful but not incremental. Table 6 shows faithfulness is acceptable — the Hebrew-native `dictalm2` is clearly more grounded than the stronger general-purpose model — but novelty is the failure. Table 7(b) shows the pattern: the second entry re-establishes the first almost point for point before adding anything. Few-shot prompting made this *monotonically worse* for both models (18.1→15.0→5.0; 10.1→8.3→5.3); in the worst condition 92% of updates were near-verbatim copies of the example's answer. For a free-text diff task, examples are imitated as *content* rather than *style* — the opposite of what we saw for classification.

End-to-end journal. Run on all 523 gold segments under predicted growth, the anti-duplication point (θ=0.342) yields 491 entries, 25 multi-session, at 0.950 purity; the best-F_1 point (θ=0.156) yields 343 entries, 80 multi-session, at 0.688 purity. This is a *demonstration artifact*: at 0.13 linking precision an end-to-end score would conflate linking with summarisation error.

Canonicalisation. Rewriting segments into neutral third-person Hebrew before embedding targets the register confound directly. On 19 protocols — with pseudo-labels from the LLM that wrote the text, so this is a mechanism rather than a measurement — it sharpened intrinsic separation (AUC $0.947 \rightarrow 0.986$) but inverted downstream: F_1 $0.114 \rightarrow 0.095$, precision $0.073 \rightarrow 0.154$, recall $0.261 \rightarrow 0.069$. Doubling precision while collapsing recall is the *desirable* direction under our cost model even though F_1 falls — F_1 is the wrong instrument here. The rewrites show why: the model compressed to a median 25% of the original length despite a prompt forbidding summarisation, dropping the identifiers of Table 7(a).

5 Conclusion

The intuitive failure mode — "Hebrew embeddings are not good enough" — is not what we found: a CPU-side representation separates same- from different-matter segments at AUC 0.954, and retrieval places the right entry in the top 10 every time. What defeats the system is the streaming decision itself, where a 6.9% base rate against a growing candidate set means an excellent false-positive rate still produces more false merges than true ones. The levers that helped were correspondingly structural — judge *distinctiveness* rather than *similarity*, and represent the accumulated matter rather than its first appearance — while larger models in the deciding role did not help at all. Finally, refitting those levers under one protocol changed what we could claim for one of them: an ablation table means little unless its rows share a fitting protocol.

Limitations

Only 36 repeat events: recall at FMR$\leq$ 5% moves in 2.8-point increments, so large effects are safe while fine rankings are not. Segmentation is gold, so an automatic system adds its own error; all results come from one templated committee; and log-writing is scored by an unvalidated 3B judge.

Statistic	Value
Protocols (meetings)	213
Protocols manually segmented	211
Protocol-level canonical topics	151
Gold segments (subject spans)	523
Distinct subject labels	487
Recurring matters ($\geq$ 2 appearances)	22
Repeat events (decidable links)	36
Base rate of repeats among arrivals	6.9%

Table 1: Data statistics. Real-world Hebrew parliamentary transcript, 25th Knesset Finance Committee, 2022-11 to 2023-10. *Repeat events* are the only arrivals with a correct link available, and their scarcity drives every result that follows.

Representation	ARI	AUC
e5, no ABTT (k=0)	0.124	0.857
e5 + ABTT (k=1)	**0.657**	**0.954**
TF-IDF	0.382	0.874
AlephBERT + ABTT (k=2)	0.265	0.854

Table 2: Segment representation on the gold set. ARI is agglomerative clustering against gold subject labels; AUC is same/different-matter separability over segment pairs.

Protocol	AUC	$\mathbf{F_1}$	P	R	FMR
Pairwise (oracle)	0.954	**0.619**	0.500	0.811	0.038
Streaming (real)	—	**0.125**	0.077	0.333	0.296

Table 3: The separability/decision gap. Identical embeddings, identical gold data, identical metric; only the decision setting changes.

Method	$\mathbf{F_1}$	R@FMR5
matched: centering, honest refit, oracle growth		
First span + threshold	0.086	0%
Centroid + threshold	0.087	0%
Centroid + **margin**	**0.190**	**8.6%**
LLM as classifier		
Best verifier (single snippet)	0.111	$\approx$0%
Verifier, full timeline	0.068	0%

Table 4: What closes the gap. **R@FMR5** is recall at the anti-duplication point (FMR$\leq$ 5%). The first block varies *one lever at a time* with the transform, its fitting protocol and the growth model held fixed (`honest_levers.json`); only the margin converts into F_1. Under this protocol the threshold rule degenerates — both threshold rows peak at recall 1.0 with precision $\approx$0.045 — so the centroid's benefit is not separable here; it is visible on the batch-fit ABTT-k1 path, where it lifts recall 32%$\rightarrow$89%.

Transform	Batch (leaky)	Honest refit
Whitening (50-dim)	0.250	0.094
Centering	0.175	0.190
ABTT-k1	0.200	0.162

Table 5: Look-ahead leak 1: the fitting protocol as an experimental variable (best F_1, streaming linking). Transforms needing many samples cannot be fitted from an early prefix and lose most of their apparent advantage.

	`dictalm2`		`qwen2.5-7b`	
Prompt	**Faith.**	**Nov.**	**Faith.**	**Nov.**
Zero-shot	**80.7**	**18.1**	63.4	**10.1**
+ example (real)	71.1	15.0	63.1	8.3
+ example (fictional)	53.6	5.0	**69.2**	5.3

Table 6: Context-conditioned log writing on 22 gold chains (58 entries), scored 0–100 by `Qwen2.5-3B-Instruct`. Faithfulness is acceptable; novelty is the failure, and few-shot examples make it monotonically worse for both generators.

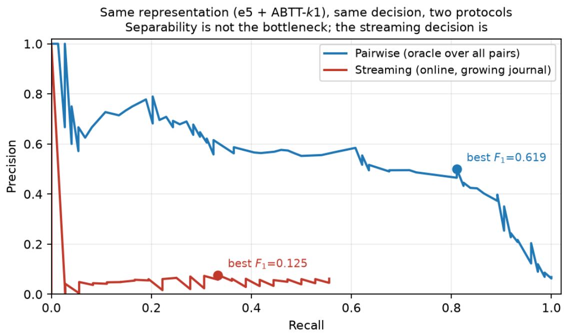

Figure 1: Precision–recall for one representation under both evaluation protocols. The oracle frontier sustains precision near 0.6; the streaming frontier never exceeds $\approx$0.08. The area between them is attributable to the decision setting, not the embedding.

(a) Why the decision fails: the register confound	*gold repeat event 8; e5+ABTT-k1 cosine*
Query 2022-12-13, Energy Ministry	"Ministry of Energy. *Request no. 177* is intended to fund 559 million shekels in commitment authority and 4 staff positions…"
True continuation 2022-12-06 $\cos = \mathbf{0.266}$	"Request approved. *Request no. 177*: Ministry of Energy. The request concerns the Ministry of Energy budget line…"
Top distractor 2022-12-06, Transport $\cos = \mathbf{0.448}$	"…they are very nice people, they smile, and they bring the request again in the same way. Ministry of Transport…"
The true match shares the ministry ***and the request number****; the distractor shares only bureaucratic phrasing, and still outranks it by* 0.18*. This is what the distinctiveness margin and the fingerprint extractor each attack.*	

(b) Why log writing fails: recap-then-append	*chain "rising milk prices", dictalm2, zero-shot*
Entry 1 2022-11-28 (opening)	"…controlled milk prices have not changed in a decade, but input costs rose, so an agreement was reached with the government on a moderate 4.8% rise. The chair, Moshe Gafni, asked why Israel does not subsidise dairy farmers as in Europe and the US, where there is **no VAT on dairy**, and why **consumers must absorb the increase** instead of the state helping the industry."
Entry 2 2022-11-28 (*should be the increment*)	"The chair, Moshe Gafni, stressed the need to protect dairy farmers and prevent shifting the burden to consumers. He recalled the difference between state support elsewhere, where **there is no VAT on dairy**, and Israel, where **consumers must absorb the increase**. Gafni added that the dairies must not exploit the situation to raise *unregulated* prices, and that the whole chain must be handled properly."
Bold marks content already present in entry 1. The genuinely new material is one clause — unregulated prices and the full supply chain — wrapped in a restatement of the previous entry. Novelty scores 18.1/100 while faithfulness scores 80.7: the model is truthful about the record and unable to emit only the diff.	

Table 7: Qualitative examples behind the two central failures. Hebrew source text translated by the authors; ellipses mark elision, emphasis added. (a) is the retrieval confound of Section 4; (b) is the novelty failure of Table 6.

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